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The resting-state brain activity signatures for addictive disorders

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Highlights

- There are shared or distinct brain regions across different addictive disorders
- Addiction increased activity in the striatum and SMA
- Addiction decreased activity in ACC and vmPFC
- Altered neural activity associated with dopamine receptor signaling pathway

Context and significance

This study aimed to uncover common and unique patterns of abnormal brain activity in individuals with substance use disorder (SUD) and behavioral addiction (BEA). Understanding these patterns is crucial for addressing addiction, a persistent condition with complex neural mechanisms. The findings reveal shared alterations in striatum and motor cortex activities, as well as reduced activity in the anterior cingulate cortex and ventromedial prefrontal cortex for both SUD and BEA. These results highlight the importance of targeted neuromodulation approaches. In

simplified terms, the study explores brain activity in addiction, emphasizing shared and distinct patterns in SUD and BEA and providing insights for targeted interventions.

Summary

Background

Addiction is a chronic and relapsing brain disorder. Despite numerous neuroimaging and neurophysiological studies on individuals with substance use disorder (SUD) or behavioral addiction (BEA), currently a clear neural activity signature for the addicted brain is lacking.

Methods

We first performed systemic coordinate-based meta-analysis and partial least-squares regression to identify shared or distinct brain regions across multiple addictive disorders, with abnormal resting-state activity in SUD and BEA based on 46 studies (55 contrasts), including regional homogeneity (ReHo) and low-frequency fluctuation amplitude (ALFF) or fractional ALFF. We then combined Neurosynth, postmortem gene expression, and receptor/transporter distribution data to uncover the potential molecular mechanisms underlying these neural activity signatures.

Findings

The overall comparison between addiction cohorts and healthy subjects indicated significantly increased ReHo and ALFF in the right striatum (putamen) and bilateral supplementary motor area, as well as decreased ReHo and ALFF in the bilateral anterior cingulate cortex and ventral medial prefrontal cortex, in the addiction group. On the other hand, neural activity in cingulate cortex, ventral medial prefrontal cortex, and orbitofrontal cortex differed between SUD and BEA subjects. Using molecular analyses, the altered resting activity recapitulated the spatial distribution of dopaminergic, GABAergic, and acetylcholine system in SUD, while this also includes the serotonergic system in BEA.

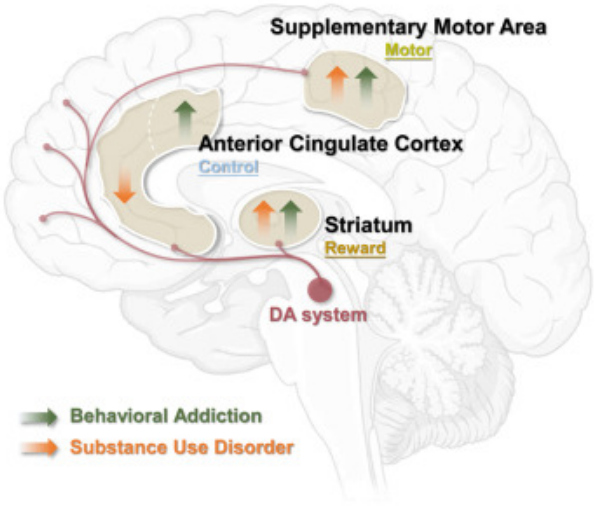
Conclusions

These results indicate both common and distinctive neural substrates underlying SUD and BEA, which validates and supports targeted neuromodulation against addiction.

Funding

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Graphical abstract



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Keywords

resting state; neural activity; substance use disorder; behavioral addiction; dopamine; serotonin

CAT scale

Translation to patients

Introduction

Addiction is a chronic relapsing brain disease characterized by compulsive seeking and either using addictive substances or conducting addictive behaviors (such as pathological gambling) despite adverse consequences.[1] This aberrant behavior is rooted in deviations of brain activity,[2] receptors,[3] and/or gene expression.[4,5] The imbalanced brain activity among diverse cortical and subcortical brain regions is considered as a core neurophysiological feature of addiction.[6] Neurobiological findings indicated that the subcortical system,[7] such as the ventral tegmental area, nucleus accumbens, and dorsal striatum (putamen, caudate), is involved in motivation toward drug use or addictive behaviors,[8] through dopamine[3] and mu receptor[9] signaling, while the prefrontal cortical system, such as the ventral medial prefrontal cortex (vmPFC), orbitofrontal cortex (OFC), and anterior cingulate cortex (ACC), is important in executive control over drug seeking,[10] cue reactivity,[11,12] and development into compulsive drug use.[13] In addition, the motor cortical system is implicated in addictive behavioral responses,[14] such as motor impulsivity regulation in addiction.[15]

However, this knowledge is derived from various studies covering different types of substance use disorder (SUD) and behavioral addiction (BEA), therefore showing high heterogeneity. Currently, we still lack unified understanding of the neurophysiological signature of addiction.

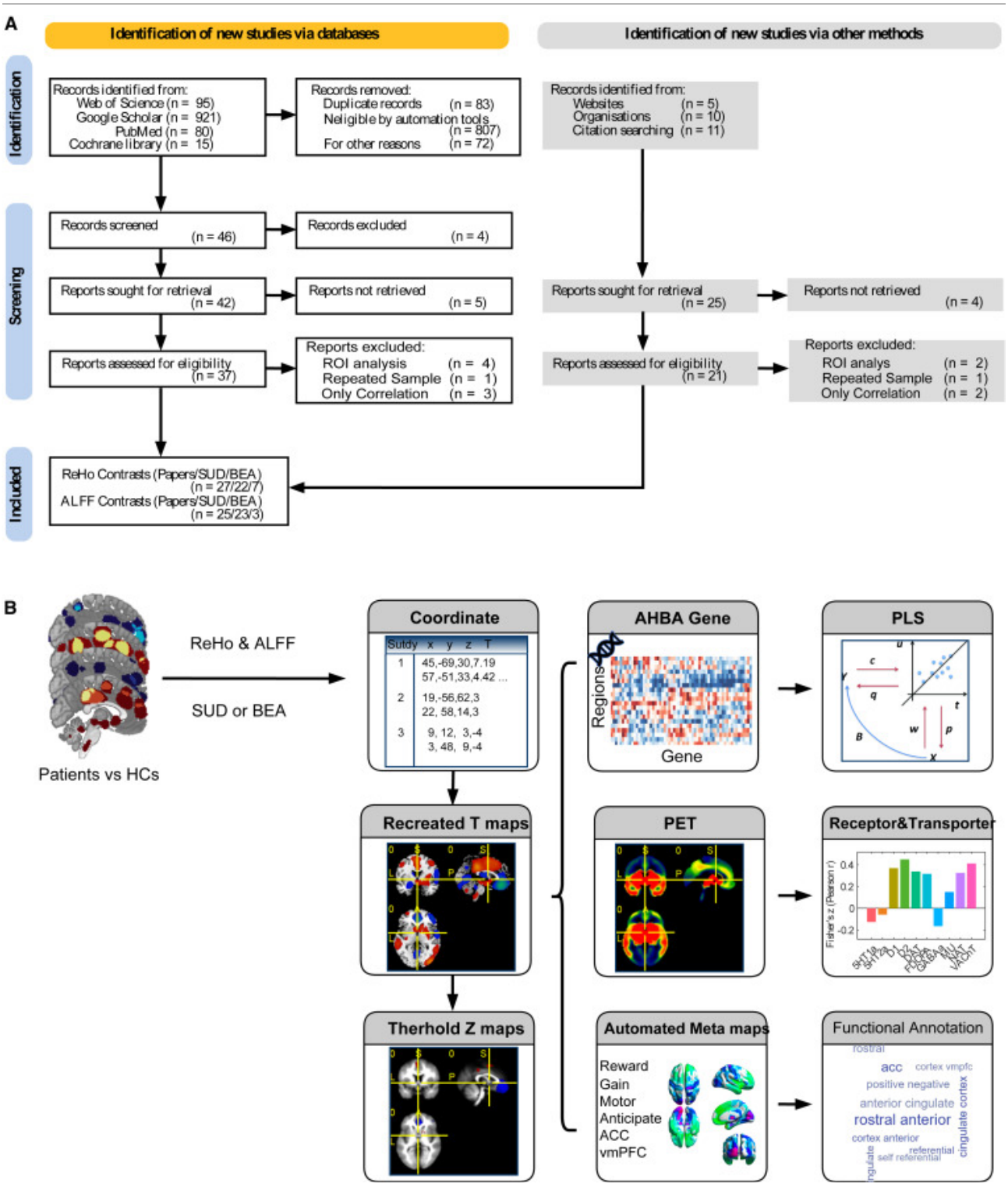
Resting-state brain activity[16] provides valuable understanding for brain disorders at system level and has been considered as an important physiological index for brain functioning, which better reflects neuroplasticity changes in brain diseases[17] when compared to structural imaging.[18,19,20] Due to its extensive generalizability across different laboratory settings, resting-state brain activity datasets are more comparable to each other and therefore suitable for meta-analysis purposes when compared to task-state datasets[21] (often with different task types and task state contrasts). Notably, two complementary metrics of resting-state functional MRI (fMRI), namely regional homogeneity (ReHo)[22] and amplitude of low-frequency fluctuations (ALFF),[23] have been shown to serve as a signature for psychiatric diseases (e.g., schizophrenia[24]). ReHo represents the temporal oscillation synchronization of fMRI signals between neighboring voxels, which is regulated by local functional connectivity.[25] Fractional ALFF (fALFF) represents the absolute and proportional intensities, respectively, of the fMRI signal, reflecting spontaneous local brain activity.[26] Multimodal evidence has indicated that resting-state brain activity in functional networks correlates with coordinated activity of dozens of genes related to ion-channel activity and synaptic function.[27] Therefore, ReHo and ALFF, as resting-state brain activity indices, are valuable for prognostic evaluation, clinical stratification, and mechanistic understanding of addiction.

Several coordinate-based (CB) neuroimaging meta-analyses have reported neuroadaptations in addiction from the perspective of resting state,[28] task state,[29] and structural,[30] or have revealed differences between addiction types. However, a common understanding of alterations of resting-state brain activity signature for addiction is still lacking. One remaining obstacle is the heterogeneous brain regions and heterogeneity in methodology (e.g., diversity in scanner settings, participant recruitment, data collection, and data analysis protocols) that further contribute to divergent neuroimaging findings of addiction.[31] The present study performed a full CB meta-analysis to identify potential common and distinct resting-state brain activity signatures in individuals with SUD and/or BEA. The involvement of gene expression and synaptic/receptor distribution are analyzed to examine their contribution to these alterations in neural activity. The findings will offer support for neuromodulation therapies targeting aberrant neural activity in the addicted brain.

Results

Clinical and demographic characteristics

Based on the inclusion and exclusion criteria, the database we generated included 46 articles (55 contrasts), 6 of which reported both ALFF and ReHo results. The ReHo dataset included 25 articles (26 contrasts) with 941 (173 female) participants with addictive disorders and 906 (164 female) healthy controls (HCs). The ALFF dataset included 27 articles (29 contrasts) with 979 (148 female) participants with addictive disorders and 967 (174 female) HCs. Most participants with addictive disorders were male (81.55%, 1,335/1,637) (Figure 1A). All the studies reported comparison(s) between participants with addictive disorders and matched HCs. The clinical characteristics and other details of the included studies are shown in Tables 1 and S7.



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Figure 1. Flow diagrams of this study

(A) An adapted preferred reporting items for systematic reviews and meta-analyses flow diagram. There were no previous meta-analyses on this topic, regional homogeneity (ReHo) or amplitude of low-frequency fluctuation (ALFF) differences between patients with addiction (substance use disorder or behavioral addiction) and healthy controls (HCs), so we identified studies by keywords and MeSH in PubMed and Web of Science databases and other methods. Finally, we included 46 studies, of which 26 were ReHo studies (1 study has two contrasts), 29 were ALFF studies (2 studies have two contrasts), and 6 included both ReHo and ALFF contrasts. ROI, region of interest.

(B) Main statistical pathway of this study. We reconstructed a t-statistical map of each contrast between patients with addiction and HCs on spontaneous activity according to the d-SDM method,[32] followed by partial least squares (PLS) regression[33] to screen for expressed genes with abnormal spontaneous activity. Spatial correlation was then used to test the relationship between such spontaneous activation differences and addiction-related receptors or transporters.[34] Finally, we searched for the cognitive term associated with this abnormal spontaneous activity through the Neurosynth database and shown by word cloud.[35]

Table1. Summary of recent whole-brain analysis resting-state fMRI study of addiction

	Addiction group							**HC grou**
Studies/ contrasts	**Average age**	**Addiction type**	**Addiction diagnostic**	**Addiction scale/s**	**Scale/ s scores**	**Addiction years**	**Abstinent time before screen**	**Sample male/ female**
ReHo								
Liu et al.[36]	20.5	internet	YDQ+	YDQ	≥5	NaN	did not report	11/8
Qiu et al.[37]	36.4	heroin	DSM-IV	NaN	NaN	9.35	methadone maintained	26/5
Yu et al.[38]	40.4	nicotine	DSM-IV	FTND	7.19	21.1	no abstinence	16/0
Dong et al.[39]	24.4	internet gaming	YIAT+	YIAT	>80	NaN	no abstinence	15/0
Liao et al.[40]	26.6	ketamine	DSM-IV	NaN	NaN	3.43	at least 48 h	33/8
Tang et al.[41]	27.1	nicotine	DSM-IV	cravings	6.41	10.2	12 h	37/8
Qiu et al.[42]	24.52	cough syrups	DSM-IV	NaN	NaN	5.08	no abstinence	28/2
Denier et al.[43] A	40.9	heroin	ICD-10	NaN	NaN	20.6	received diacetylmorphine 60 min	21/8
Kim et al.[44] A	23.51	internet gaming	DSM-5	YIAT	75.81	NaN	did not report	16/0
Kim et al.[44] B	27	alcohol	DSM-IV	AUDIT-K	27	NaN	at least 2 weeks	14/0
Wu et al.[45]	46.6	nicotine	DSM-IV	FTND	8.87	≥5	no abstinence	24/7

	Addiction group							HC grou
Studies/ contrasts	Average age	Addiction type	Addiction diagnostic	Addiction scale/s	Scale/ s scores	Addiction years	Abstinent time before screen	Sample male/ female
Liu etal.[46] A	46.25	betel quid	NaN	BQDS	10	20.6	the scanning day	24/9
Chen and Mo [47]	35.49	nicotine	DSM-IV	FTND	5.1	≥5	no abstinence	14/0
Tu etal.[48]	48.55	alcohol	DSM-IV	SADQ, AUDIT	20.34, 23.83	27.93	first-time visitors	20/9
Weng etal.[49] A	34.3	betel quid	NaN	BNDS	28.4	14.46	did not report	16/0
Zhang etal.[50]	22.1	methamphetamine	ICD-10	NaN	NaN	5.2	some days	9/8
Yu etal.[51]	33.35	methamphetamine	NaN	NaN	NaN	5.033	did not report	45/0
Weng etal.[52]	29.34	nicotine	NaN	FTND	4.65	9.025	4.99 h	67/0
Xie etal.[53]	28.1	methamphetamine	DSM-IV	NaN	NaN	5	38.5 days	59/11
Yang etal.[54]	34.91	methamphetamine	DSM-IV	NaN	NaN	4.03	>15 days	24/0
Breno etal.[55] A	30.55	cocaine	DSM-IV	CSSA, ASI-6	15.80, 50.5	12.55	>14 days	20/20
Deng etal.[56] A	39.5	alcohol	DSM-IV	AUDIT	20.13	20.79	5–12 days	68/0
Nie etal.[57]	27.35	methamphetamine	NaN	NaN	NaN	4.77	>12 days	51/46
Wen etal.[58]	23.33	nicotine	NaN	FTND	6.32	5.23	~1 h	32/24
Xue etal.[59]	33.23	heroin	DSM-IV	NaN	NaN	6.36	12 h	26/0
Zhang etal.[60]	31.86	nicotine	DSM-V	FTND	3.67	12.61	did not report	52/0
ALFF/fALFF								
Jiang etal.[61]	35.525	heroin	DSM-IV	NaN	NaN	10.83	6 or 7 days	20/4

	Addiction group							HC group
Studies/ contrasts	Average age	Addiction type	Addiction diagnostic	Addiction scale/s	Scale/s scores	Addiction years	Abstinent time before screen	Sample male/ female
Orr etal.[62] f	16.3	cannabis	DSM-IV	NaN	NaN	3.56	the night	17/0
Wang etal.[63]	34.1	heroin	DSM-IV	NaN	NaN	6.79	1 week	17/0
Yuan etal.[64]	19.45	internet gaming	YDQ+	YDQ	NaN	2.9	did not report	12/6
Chu etal.[65] f	30	nicotine	ICD-10	NaN	NaN	>5	did not report	9/0
Ide etal.[66] f	38.85	cocaine	DSM-IV	NaN	NaN	18	did not report	55/29
Denier etal.[43] B f	40.9	heroin	ICD-10	NaN	NaN	20.6	received diacetylmorphine 60 min	21/8
Lin etal.[67] f	22.24	internet gaming	DSM-5	YIAT	>80	NaN	no abstinence	26/0
Feng etal.[68] f	20.6	nicotine	DSM-5	QSU, FTND	26, 6	4.9	did not report	27/0
Liu etal.[46] B	46.25	betel quid	BQDS+	BQDS	10	20.6	on the scanning day	33/0
Park etal.[69]	23.7	internet gaming	DSM-5	YIAT	62.4	NaN	did not report	24/0
Qiu etal.[70]	24.5	codeine-containing cough syrups	DSM-IV	NaN	NaN	4.96	denied use in the month	14/0
Wang etal.[71] f	38.4	nicotine	DSM-IV	FTND	5.2	19	no abstinence	55/0
Han etal.[72]	16.9	internet gaming	YDQ+	CIAS	71.88	NaN	did not report	26/0
Liu etal.[73]	48.55	alcohol	DSM-IV	SADQ, AUDIT	20.34, 23.83	27.93	did not report	20/9
Sun etal.[74]	21.39	internet gaming	YDQ+	CIAS	74.39	2.18	did not report	30/23
Weng etal.[49] A	34.3	betel quid	NaN	BNDS	28.4	14.46	did not report	16/0

	Addiction group							HC grou
Studies/ contrasts	Average age	Addiction type	Addiction diagnostic	Addiction scale/s	Scale/ s scores	Addiction years	Abstinent time before screen	Sample male/ female
Wang etal.[75] f	40.02	nicotine	NaN	FTND	4.72	20.28	did not report	19/1
Horvath etal.[76]	22.75	smartphone	SAS-SV	SPAI	57.2	NaN	did not report	7/15
Liu etal.[77] A	29.9	methamphetamine	DSM-V	NaN	NaN	2.425	did not report	21/0
Liu etal.[77] B	29.85	heroin	DSM-V	NaN	NaN	3.575	did not report	21/0
Luo etal.[78]	42.375	heroin	DSM-V	NaN	NaN	15.65	26.73 days	36/15
Qiu etal.[79] A	22.585	nicotine	NaN	FTND	6.28	5.18	did not report	31/15
Qiu etal.[79] B	20.975	internet gaming	NaN	CIAS	72.68	NA	did not report	26/12
Hirjak etal.[80]	23.8	cannabis	NaN	CUDIT	17.63	4.56	2v4 h before	24/0
Wen etal.[81]	34.15	nicotine	NaN	FTND	6.7	14.4	30–45 min smoke	30/0
Weng etal.[52] f	29.34	nicotine	NaN	FTND	4.65	9.025	4.99 h	67/0
Xie etal.[53] f	28.1	methamphetamine	DSM-IV	NaN	NaN	5	38.5 days	59/11
Deng etal.[56] B f	39.5	alcohol	DSM-IV	AUDIT	20.13	20.79	5–12 days	68/0

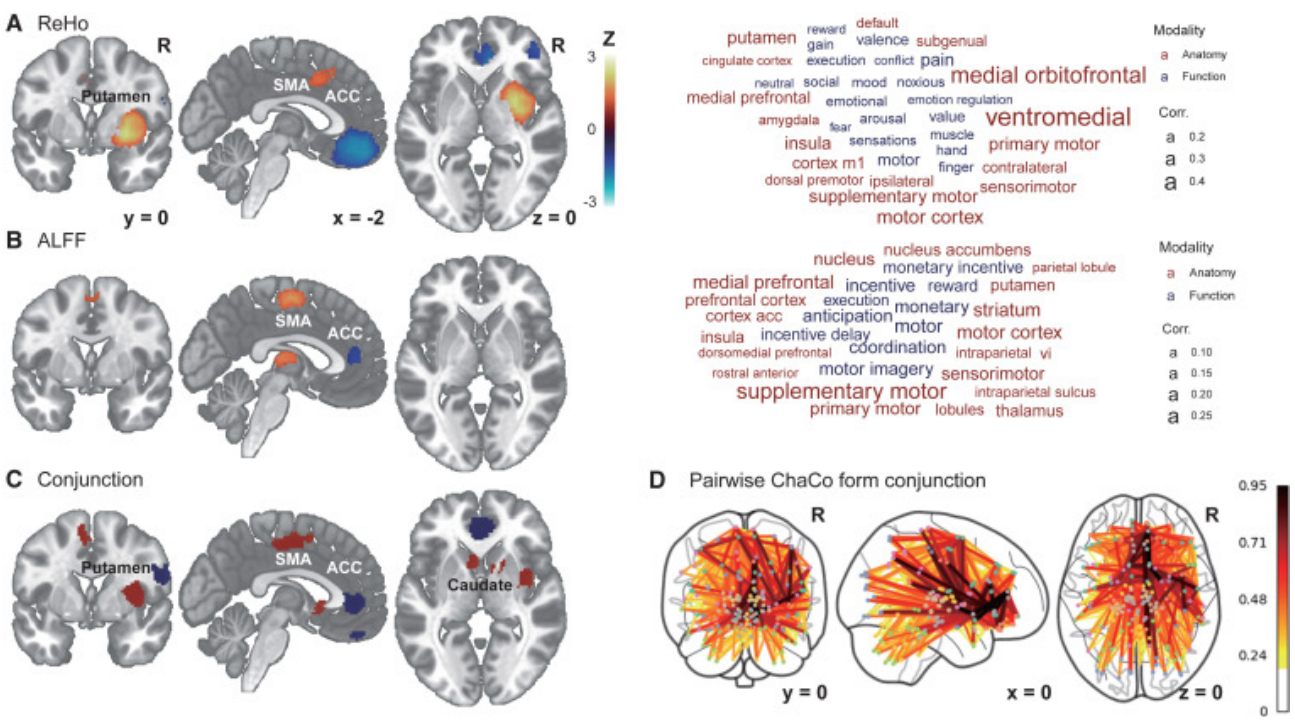
Most of this study used filtering of 0.01–0.08 Hz, and REST software calculated the ReHo and ALFF.

YDQ, Young's Diagnostic Questionnaire for Internet Addiction; +, additional diagnostic conditions in addition to addiction scale/s; FTND, Fagerström Test for Nicotine Dependence; DSM-IV, Diagnostic and Statistical Manual of Mental Disorders, 4th edition; YIAT, Young's Internet Addiction Test; ICD-10, International Classification of Diseases—10; BQDS, betel quid dependence scores; BNDS, Betel Nut Dependency Scale; CIDI-SF, World Health Organization Composite International Diagnostic Interview Short Form; QSU, 10-item brief Questionnaire on Smoking Urges; CIAS, Chen Internet Addiction Scale; SADQ, Severity of Alcohol Dependence Questionnaire; AUDIT, Alcohol Use Disorders Identification Test; SAS-SV, short version of the Smartphone Addiction Scale; SPAI, Smartphone Addiction Inventory; CSSA, Cocaine Selective Severity Assessment; ASI-6, Addiction Severity Index; CUDIT, Cannabis Use Disorder Identification Test; FWHM, full width at half maximum; NaN, null of data;

MCC, multiple comparison correction; FDR, false discovery rate; Uncorr., uncorrected for multiple comparisons; FWE, family-wise error; Corr., corrected for multiple comparisons; GRFT, Gaussian Random Field Theory; AlphaSim, AlphaSim correction; TFCE, threshold free cluster enhancement; f, fALFF study; A, first contrast condition; B, second contrast condition.

Deviations of ReHo and ALFF in the addiction group

To obtain a common pattern of spontaneous brain activity, we performed a meta-analysis of ReHo and ALFF separately and obtained the intersection of these two indices. Compared to the HC group, higher ReHo was observed in the right putamen, right insula (Brodmann area [BA] 48), bilateral supplementary motor area (SMA; BA 6), bilateral precentral gyrus (BA 4), and right inferior temporal gyrus (BA 37) in the addiction group. In comparison, lower ReHo was observed in the bilateral ACC (BA 10, 11), bilateral vmPFC (BA 11, 10), right middle frontal gyrus (inferior frontal gyrus [IFG]; BA 45, 46), right superior frontal gyrus (dorsolateral prefrontal cortex; BA 9), and right postcentral gyrus (BA 4, 43) in the addiction group (TableS3 and Figure2A). Compared to the HC group, higher ALFF/fALFF was observed in bilateral SMA (BA 6) and bilateral thalamus, while lower ALFF/fALFF was observed in bilateral ACC (BA 32) and vmPFC (BA10) in the addiction group (TableS3 and Figure2B). The conjunction map of higher ReHo and higher ALFF included right dorsolateral striatum (putamen, insula; BA 48), bilateral SMA (BA 6), and left striatum (caudate nucleus; BA 25). The conjunction map of lower ReHo and lower ALFF included bilateral ACC (BA 32), vmPFC (BA 10), and right postcentral gyrus (BA 48). No overlap of higher ReHo and lower ALFF, or vice versa, was found (TableS3 and Figure2C).



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Figure2. Common altered resting brain activity in addiction

(A) Brain regions with regional homogeneity (ReHo) deviation in patients with addiction. Main differences between patients with addiction and HCs in the coordinate-based meta-analysis (also in B and C). Compared with HCs, addiction patients show higher ReHo in the right putamen and right

insula (Brodmann area [BA] 48), bilateral supplementary motor area (SMA; BA 6), bilateral precentral gyrus (BA 4), and right inferior temporal gyrus (BA 37), and lower ReHo in the bilateral anterior cingulate cortex (ACC; BA 10, 11), bilateral ventromedial prefrontal cortex (vmPFC; BA 11, 10), right middle frontal gyrus (inferior frontal gyrus, BA 45, 46), right superior frontal gyrus (dorsolateral prefrontal cortex, BA 9), and right postcentral gyrus (BA 4, 43) in addiction patients ($p<0.005$, uncorrected for false discovery rate [FDR]; peak height Z value >1; extent threshold $k>50$ voxels). Corr., correlation.

(B) Amplitude of low-frequency fluctuation (ALFF) differences between patients with addiction and HCs in the coordinate-based meta-analysis. Compared with HCs, addiction patients displayed significantly higher ALFF/fALFF in bilateral SMA (BA 6) and bilateral thalamus, and lower ALFF/fALFF in bilateral ACC (BA 32) and ventromedial prefrontal cortex (vmPFC, BA10) in addiction patients ($p<0.005$, uncorrected for FDR; peak height Z value >1; extent threshold $k>50$ voxels).

(C) Conjunction of ALFF and ReHo differences between patients with addiction and HCs in the coordinate-based meta-analysis. The hot color (red) represents ReHo increase converged with ALFF increase in the right dorsolateral striatum (putamen, insula, BA 48), bilateral SMA (BA6), and left striatum (caudate nucleus, BA 25). The cold color (blue) represents ReHo decrease converged with ALFF decrease in bilateral ACC (BA 32), vmPFC (BA 10), and right postcentral gyrus (BA 48). The probability is 0.0025, the peak height threshold is 0.00025, and the extent threshold is >50. The tolerable p value is 0.01.

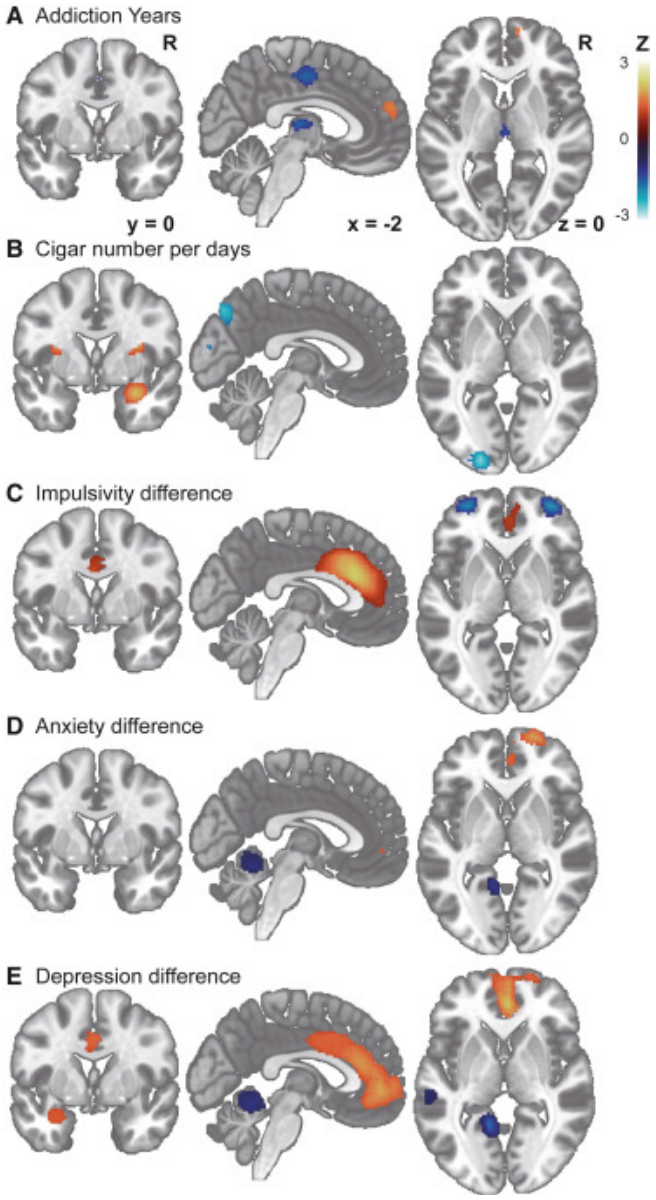
(D) The estimated change in connectivity (ChaCo) score (0–1) between 116Gy matter regions (AAL1) primarily on structural connections between the frontal and parietal lobes. The deviation of spontaneous activation was correlated with several meta-analytic anatomies and/or cognitive terms (http://neurosynth.org ↗).

Based on Neurosynth analysis (https://www.neurosynth.org/ ↗), the altered spontaneous activity was correlated with several anatomic regions and/or sensorimotor and cognitive functions (Figures 2A and 2B). The altered ReHo was correlated with sensorimotor and cognitive functions related to reward decision making, while the altered ALFF was correlated with motor and reward anticipation functions. Based on the analysis that links the functional connectivity of gray matter regions to their white matter fiber connections, the difference between addiction patients and the HC group relies primarily on structural connections between the frontal and parietal lobes, as shown in Figure 2D. Abnormal functional network mapping is also supported by abnormalities in the prefrontal lobes, ACC, and striatum to some extent (Figure S2). In addition, subgroup analyses suggest a certain level of variation in the results (Figure S3 and Table S6).

Associations of ReHo and ALFF with addiction history, personality trait, and affective measures

To explore factors related to altered ReHo and ALFF in the addiction group, we included, respectively, years of addiction, the number of cigarettes smoked per day, level of impulsivity (relative to the HC group), level of anxiety (relative to the HC group), and level of depression as predicting variables for differences in brain activity (ReHo and ALFF) in the addiction group. Meta-regression analysis revealed that the duration of addiction was associated positively with ALFF in the left middle frontal

gyrus and right superior frontal gyrus and negatively with ALFF in the bilateral ACC and thalamus (Figure 3A). Number of cigarettes smoked per day was correlated positively with ALFF in the right amygdala, right inferior temporal gyrus, right putamen, left caudate nucleus, and left stratum and negatively with ALFF in the left optic radiations and left precuneus (Figure 3B). Impulsivity was associated positively with ALFF in the bilateral ACC and left middle frontal gyrus and negatively with ALFF in the bilateral medial OFC (Figure 3C). Level of anxiety was correlated positively with ALFF in the right superior frontal gyrus, right inferior temporal gyrus, and left fusiform gyrus and negatively with ALFF in the left fusiform gyrus and cerebellum (Figure 3D). Finally, the severity of depressive symptoms was associated positively with ALFF in the bilateral ACC and left medial network and negatively with ALFF in the left cerebellum and left middle temporal gyrus (Figure 3E). More details are presented in Figure S4 and Tables S7 and S8.



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Figure 3. Deviations in ALFF predicted by clinical indicators

(A) Using the addiction year (n = 22) to predict the ALFF deviation in substance use disorder (SUD).

(B) Using the number of cigarettes per day (n = 15) to predict the ALFF deviation in SUD.

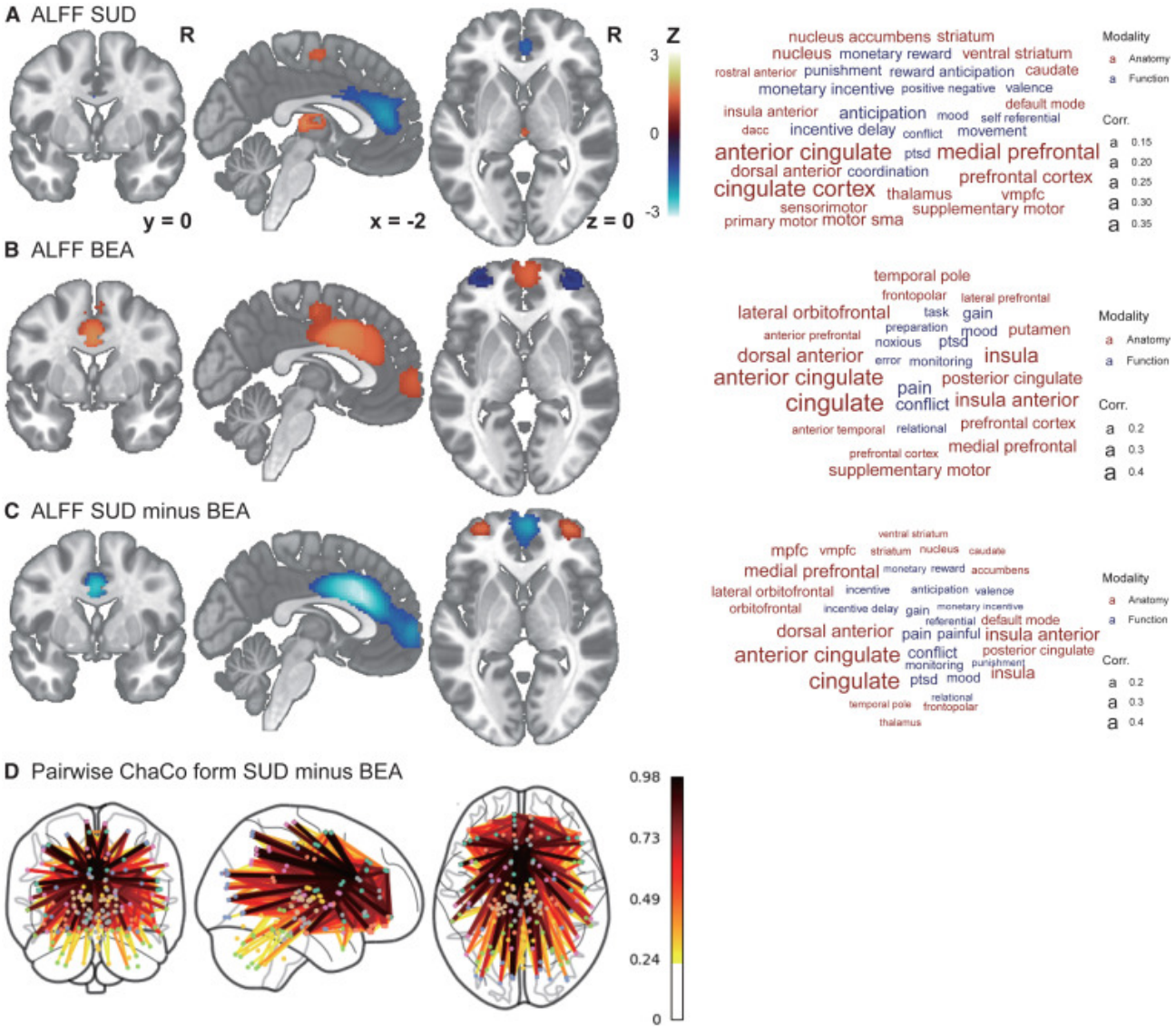
(C) Using the impulsivity difference (the independent t value from the comparison between SUD and HC, n= 8) to predict the ALFF deviation in SUD.

(D) Using the anxiety difference (n= 9) to predict the ALFF deviation in SUD.

(E) Using the depression difference (n= 11) to predict the ALFF deviation in SUD.

Comparisons between SUD and BEA

To explore differences in spontaneous brain activity in SUD and BEA, we further analyzed resting-state activity, gene expression profiles, transmitter distribution, and functional terms. ALFF in participants with SUD was higher in the bilateral OFC (BA 47, 11) and lower in the bilateral ACC (BA 32, 24), bilateral vmPFC (BA 10), and middle cingulate cortex (MCC; BA 24), compared to those with BEA (TableS9 and Figure4C). These differences were mainly driven by the lower ALFF in the ACC and vmPFC (BA 10) for the SUD group (Figure4A) compared with their HCs and higher ALFF in the MCC (BA 24, 32) and lower ALFF in the bilateral OFC (BA 47, 11) for the BEA group (Figure4B) compared with their HCs (TableS9).



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Figure4. Distinct resting brain activity between substance use disorder and behavioral addiction
For ALFF, there are relatively sufficient substance use disorders (SUDs) and behavioral addictions (BEAs) to compare.

(A) Group comparison between SUD and BEA. Compared with patients with BEA, the patients with SUD have the ALFF higher in the bilateral lateral orbitofrontal cortex (BA 47) and lower in the bilateral ventromedial prefrontal cortex (BA 10), anterior cingulate cortex (BA 32), and left cerebellum (BA 19) ($p<0.005$, uncorrected for FDR; peak height Z value >1; extent threshold $k>50$ voxels).

(B) ALFF altered in patients with SUD shows higher ALFF in the bilateral orbitofrontal cortex (OFC; BA 47, 11) and lower ALFF in bilateral ACC (BA 32, 24), bilateral vmPFC (BA 10), and middle cingulate cortex (MCC; BA 24).

(C) ALFF altered in patients with BEA shows that increased ALFF in the bilateral medial cingulate (BA 24, 32), anterior cingulate cortex (BA 23, 23, 32), bilateral supplementary motor area (BA 6), and bilateral ventromedial prefrontal cortex (BA 10), and decreased ALFF in the bilateral lateral orbitofrontal cortex (BA 47).

(D) Estimated change in connectivity (ChaCo) score (0–1) between 116Gy matter regions (AAL1) values primarily on structural connections in the occipitofrontal fasciculus.

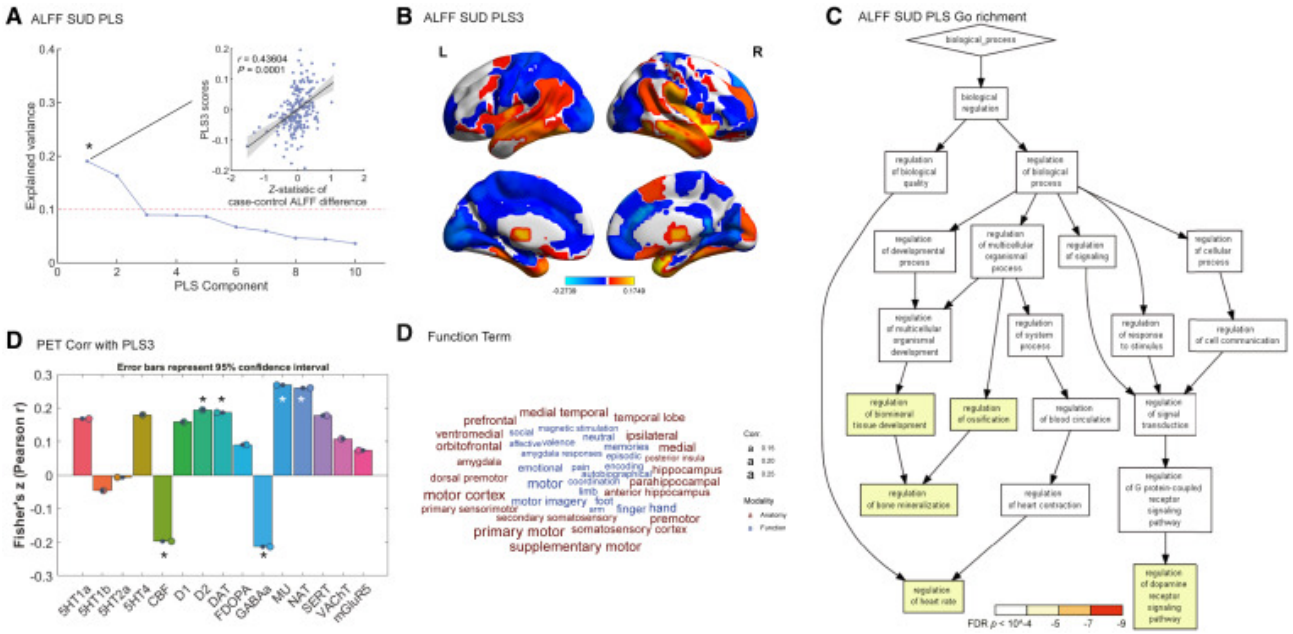
The deviation in spontaneous activation between SUD and BEA was correlated with several meta-analytic anatomies and/or cognitive terms (https://neurosynth.org/ ↗).

The meta-analysis conducted on Neurosynth showed that the altered ALFF in SUD was correlated with several cognitive functions mainly involved in monetary reward, anticipation, value, conflict, and mood processes. In BEA, the altered ALFF was associated with mood and conflict. The ALFF difference between SUD and BEA was correlated with conflict, anticipation, mood, and punishment. The differences in ALFF between the SUD and BEA groups relied primarily on structural connections between the frontal and parietal lobes through the corpus callosum and frontal-occipital fasciculus, as shown in Figure4D. In ACC subregions, receptors/transporters showed correlations with the brain activity of MCC and dorsal ACC (dACC) in SUD but not in BEA (FigureS5).

Relevance of abnormal ALFF for SUD to gene expression profiles and transmitter distribution

We attempted to obtain the relevant genes and receptors for these abnormal brain regions by performing regression and spatial correlation analyses on a gene expression database of the human brain. The third component of the partial least squares (PLS) regression (PLS3) explained 19.01% of the variance in deviation of ALFF in SUD ($p= 0.0185$, corrected for a 10,000 times permutation test for spatial autocorrelation) and showed a significant correlation with the ALFF difference ($r= 0.44$, $p= 0.0001$, Figure5A). PLS3 was characterized by a high expression mainly in the bilateral temporal lobe, SMA, inferior frontal gyrus (IFG), dorsolateral PFC (dlPFC), amygdala, and thalamus, with low expression mainly in the default network regions (Figure5B). Gene Ontology (GO) enrichment analysis showed that the PLS3-weighted enriched genes (descending arranged) were associated with the regulation of dopamine receptor signaling pathway and heart rate (false discovery rate

[FDR]-corrected p<0.05, Figure 5C). PLS3 expression in the brain was significantly positively correlated with the spatial distribution of dopamine receptor D2, dopamine transporter, μ-opioid (mu) receptor, and norepinephrine transporter; while it was negatively correlated with cerebral blood flow and $GABA_A$ receptor in the brain (p<0.005, Figure 5D). In addition, the PLS3 in SUD was correlated with several meta-analytic anatomies and/or cognitive terms mainly involved in motor, memory, and affective behavior (Figure 5E)



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Figure 5. Association of addiction-related ALFF deviations in SUD with receptor distribution and gene expression

(A) Partial least-squares (PLS) regression analysis of addiction-related ALFF deviations in SUD by gene expression identified genes show that the third components have significant expression in the Y variance (the ALFF deviation in SUD) and show that the DRD2 has a very high order of variable importance in projection (14/15,633). * p < 0.05.

(B) The component (PLS3) identified a gene expression profile that was predominantly expressed in the bilateral temporal lobe, SMA, motor area IFG, dlPFC, amygdala, and thalamus. In contrast, brain areas of the default network were lowly expressed. These transcriptional profile transcriptograms were positively correlated with the inter-group z plot of the ALFF deviation in SUD (autocorrelation-corrected permutation test, Brainsmash, 10,000 times). Shading indicates 95% confidence intervals.

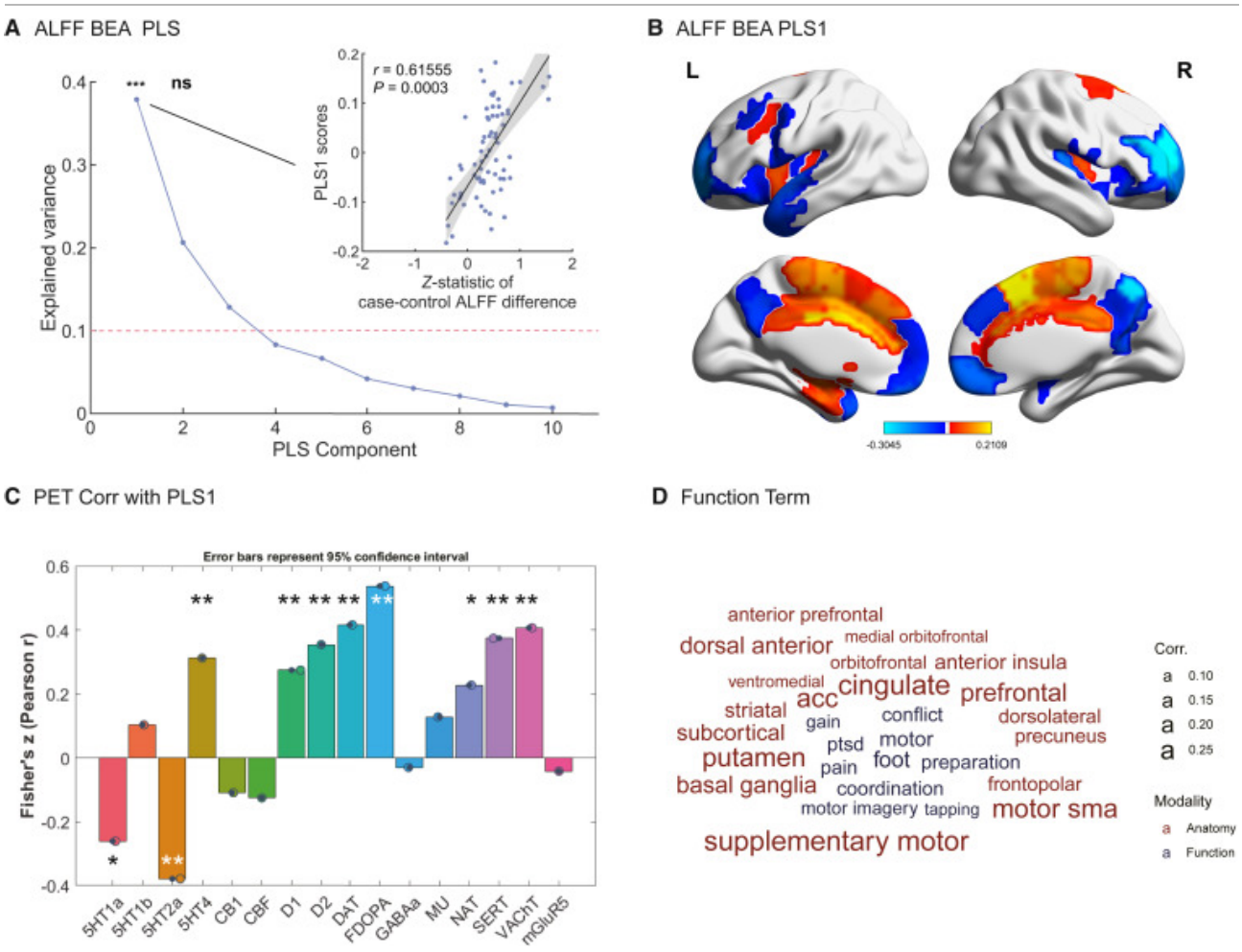
(C) Genes arranged in descending order by PLS3 weight are enriched when biological components show dopamine regulation (FDR q<0.05).

(D) PLS3 expression in the brain was significantly positively correlated with the spatial distribution of dopamine receptor D2, dopamine transporter, and norepinephrine transporter in the brain (p<0.005).

(E) Further correlation between the PLS3 in SUD and several meta-analytic anatomies and/or cognitive terms.

Relevance of abnormal ALFF for BEA to gene expression profiles and transmitter distribution

We further examined the gene expression profile changes in BEA in relation to altered neural activity. The first component of the PLS regression (PLS1) explained 37.89% of the variance in deviation of ALFF in BEA (p= 0.0332, corrected for a 10,000 times permutation test for spatial autocorrelation). It significantly correlated with the ALFF difference (r= 0.62, p= 0.0003, Figure6A). PLS1 was characterized by a high expression mainly in the midline from the ACC to posterior cingulate cortex (PCC) and insula. In contrast, it showed low expression in the dorsomedial PFC (dmPFC), vmPFC, dlPFC, and precuneus (Figure6B). GO enrichment analysis showed that enriched genes did not show any significant results (FDR-corrected p>0.05). However, PLS1 expression in the brain was significantly positively correlated with the spatial distribution of serotonin 5-HT_4 receptor agonists (5HT4), dopamine receptor D1 (DRD1), dopamine receptor D2 (DRD2), dopamine transporter (DAT), fluorodopa (F-DOPA), noradrenaline transporter (NAT), serotonin transporter (SERT), and vesicular acetylcholine transporter (VAChT) and negatively correlated with 5HT1a and 5HT2a in the brain (p<0.005, Figure6C). Neurosynth analysis showed that PLS1 in the BEA group was correlated with several anatomic regions and/or motor and cognitive functions (Figure6D).



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Figure6. Association of addiction-related ALFF deviations in BEA with receptor distribution and gene expression

(A) Partial least-squares (PLS) regression analysis of addiction-related ALFF deviations in BEA by

gene expression identified genes show that the third components have significant expression in the Y variance (the ALFF deviation in BEA) and show deficient order of variable importance in projection (13,181/15,633). The subsequent PLS and PET correlations are affected by the fact that the BEA results cannot cover the whole brain (only 78 brain regions in 268 regions of interest) and that the results are unstable. * * * $p < 0.001$; ns, not significant.

(B) The component (PLS1) identified a gene expression profile predominantly expressed in the midline from ACC to PCC, and insula. In contrast, brain areas of the dmPFC, vmPFC, dlPFC, and precuneus were lowly expressed. These transcriptional profile values were positively correlated with the inter-group z plot of the ALFF deviation in BEA (autocorrelation-corrected permutation test, Brainsmash, 10,000 times). Shading indicates 95% confidence intervals.

(C) Genes arranged in descending order by PLS1 weight is enriched during biological components and show no significant correlation with GO terms (FDR $q<0.05$).

(D) PLS1 expression in the brain was positively correlated with the spatial distribution of 5HT4, D1, D2, DAT, F-DOPA, NAT, SERT, and VAChT and negatively correlated with 5HT1a and 5HT2a ($p<0.005$).

(E) Further correlation between the PLS1 in SUD and several meta-analytic anatomies and/or cognitive terms.

Discussion

This study aimed to identify common and distinct resting-state brain activity signatures that underlie addiction in individuals with SUD and BEA using a CB neuroimaging meta-analysis approach. We used ReHo and ALFF as resting-state brain activity signatures, reflecting local brain functional synchrony and spontaneous activity intensity. We found that common resting-state brain activity abnormalities were present in SUD and BEA, mainly involving the ACC, striatum, and SMA, which were related to gene expression and dopaminergic and GABAergic neurotransmitter receptor distribution. We also found distinct resting-state brain activity abnormalities in SUD and BEA, mainly involving the ACC and OFC, which may be related to reward, executive control, and motor processes specific to each addiction type. Taken together, our findings support the Genetically Informed Neurobiology of Addiction model.[82] Understanding the neuroadaptations across addiction could facilitate the design of successful clinical trials using neuromodulation techniques to treat addiction, especially when targeting the prefrontal cortex with transcranial magnetic stimulation (TMS) or transcranial electrical stimulation.[83]

Common resting-state brain activity abnormalities in SUD and BEA

In both SUD and BEA, ReHo and ALFF were decreased in ACC, indicating reduced functional synchrony and spontaneous activity. This abnormal activity reflects the central role of ACC in addiction, including impaired cognitive control, habitual behavior formation, risk of relapse, and effectiveness of intervention.[84] The relationship between abnormal activity in ACC and addiction is discussed here in greater detail.

First, metabolic impairment of the ACC during adolescence is a risk factor for addiction in adulthood. Decreased GABA may be an important neurobiological mechanism in immature adolescent brains,

leading to a reduced ability to inhibit risky behaviors and make suboptimal decisions quickly.[85] Second, reduced ACC activation is a typical feature in addiction[86] and is consistent with the results of a previous meta-analysis,[87] with significant metabolic abnormalities in the ACC region. For example, a significant decrease in the prefrontal lobe and ACC metabolism was found in alcohol use disorder,[88] with increased AMPA receptor expression in the ACC that was positively correlated with DAT.[89] Third, activation of the ACC is a predictor of relapse in addiction, with higher activation under alcohol-cue-induced conditions in the vmPFC and ACC associated with alcohol relapse,[12] and higher Bayesian prediction errors in the right middle frontal gyrus/ACC, caudate, anterior insula, and hippocampus predicting stimulant use problems 3 years later.[90] In contrast, successful quitters had higher activation in the right dlPFC and ACC during an executive control task (e.g., Stroop task).[91] Finally, ACC activity is an indicator of efficacy of addiction interventions. Burst stimulation to the ACC via double-cone coils inhibited alcohol craving for up to 6weeks,[92] and deep TMS to the vmPFC-ACC reduced functional connectivity within regions and further reduced the number of heavy drinking days.[93] Meditation intervention for smokers increased activity in ACC and reduced smoking,[94] while synchronized fMRI-TMS reversed reward evaluation responses of ACC regions in smokers.[95] These results support ACC as a potential regulatory center and a target brain region for developing non-invasive brain stimulation (NIBS) in addiction.[96]

The interaction between the striatum and the frontal lobes plays a central role in the balance between goal-directed and habitual behaviors in both SUD and BEA.[97] The striatum showed increased ReHo in both SUD and BEA, indicating increased spontaneous activity. This aberrant activity in the brain region relevant to the reward aspect of addiction is a well-recognized finding from previous studies[98] of, e.g., nicotine dependence,[99] cocaine addiction,[100] and internet gaming disorder (IGD).[72] The dorsal striatum (putamen in humans) plays an important role in behavioral conditioning, especially habituation.[101] In HCs, overtrained habitual behavior was associated with increased putamen activity.[102] The progression of addiction from goal-orienting to more obsessive phases, such as habitual control, is accompanied by a shift in activity from the ventral striatum to the putamen.[103] Resting-state functional connectivity studies revealed compelling evidence suggesting that abnormalities in these circuits/networks underpin complex behavioral manifestations in individuals with additive disorders at different stages.[104,105] Compared with HCs, patients with BEA showed gray matter atrophy in the extended frontal cortex (ACC, vmPFC, dlPFC, and OFC), right putamen, and right SMA.[106] Extended frontal cortex abnormal activity is also shown in SUD and BEA, primarily in the form of reduced activation and decreased functional connectivity in prefrontal regions. This abnormal activity reflects impaired inhibitory control of the frontal lobes in addiction.[107] Studies using diffusion MRI have found that heroin patients have significantly reduced fractional anisotropy of the bilateral striatal-frontal circuits extending from limbic structures to the prefrontal association cortex, impairing the crista and superior longitudinal fasciculus tendon.[108] Transcranial alternating current stimulation of the frontal lobe in rats rescues cocaine-induced prefrontal hypofunction and restores flexible behavior.[109] Moreover, the inhibition-related connectivity between the anterior putamen and right insula is higher in smokers than in non-smokers.[110] In HCs, lower brain activation in the right extended frontal cortex and SMA during inhibition was associated with higher brain activation in the left ventral striatum.[111]

In SUD and BEA, the SMA is an important region for motor control and for the output of overtrained habitual behaviors.[112] Our studies have shown that SMA activity is consistently increased in addicts with ReHo and ALFF and is negatively correlated with addiction duration. The SMA contributes to

automated behavioral and motor planning, such as response initiation, selection, and inhibition, all of which play an important role in habitual control in a variety of addictive disorders.[113] In a previous study of alcohol-dependent individuals with poor initial response suppression, improvements in response time induced by modafinil were found to be associated with increased activation of the SMA (as well as the thalamus).[114] In patients with heroin-use disorders, lower activity in the left SMA during inhibitory control was associated with a shorter duration of last use and higher severity of dependence.[115] In patients with cocaine use disorder, the SMA showed higher activation during the large-reward period than during the no-reward period, and the SMA activation was positively correlated with addiction severity.[116] Again, these brain regions relevant to addiction may be used as targets of non-invasive treatment via functional connectivity.[13] TMS intervention, for example, can affect remote brain regions through functional connectivity.[117] Relative to HCs, patients with online gaming disorder (one kind of BEA) showed lower ventral striatum-SMA and higher dorsal striatum-SMA functional connectivity.[118] Repetitive TMS over the SMA improves the efficiency of inhibition control vs. effective, sustained response with increased cerebral blood flow in the left anterior SMA, left IFG, right primary motor cortex, and right IFG.[119] These results suggest that the motor area is a critical region for habitual behavior in addiction and a potential target for intervention.[120]

Distinct resting-state brain activity abnormalities in SUD and BEA

SUD showed decreased ALFF in the ACC while BEA showed increased ALFF in the ACC and decreased ALFF in OFC, indicating frontal cortex dysfunction in SUD and hyperactivity in BEA. These abnormalities may be related to different cognitive control processes in SUD and BEA,[121] such as impaired decision-making and increased impulsivity in SUD and increased compulsivity and rigidity in BEA. These deviations in brain activity were associated with gene expression and receptor distribution patterns. Specifically, while patients with SUD displayed decreased ALFF in the ACC and increased ALFF in the striatum and SMA, those with BEA did not exhibit these changes. Moreover, our analysis indicated that these alterations in brain activity among those with SUD were linked to elevated dopamine receptor density and decreased $GABA_A$ receptor density levels. These deviations may be based on neurotransmitters and are important manifestations of maintaining addiction as well as possible causes of addiction.

In both SUD and BEA, the different subregions of the ACC show different patterns of abnormal activity. The ACC includes several subregions that dominate projections to major brain systems associated with executive control (dorsal prefrontal cortex, ventral lateral prefrontal cortex, frontal pole, and parietal cortex), motor control (motor cortex, supplementary motor cortex, premotor cortex, and spinal cord), and emotion (amygdala, hypothalamus, and vmPFC).[122] First, the area of impairment varies; second, the severity of the impairment varies; and finally, the consequences of the impairment vary. Across studies, dimensions, and clinical populations, deactivation of the rostral portion of the ACC (rACC) and the prefrontal cortex (vmPFC) is the most consistent finding in SUD, which plays an important role in regulation of impaired emotion.[123] In opioid users, activity in the rACC is reduced during emotional processing, and rACC gray matter volume correlates with lower frequency of use and emotional awareness. In contrast, activation of the ventral ACC (vACC) is associated with an increase in emotionally salient tasks.[124] In BEA individuals, such as pathological gamblers, rACC activation is associated with almost missing out (a losing situation that is [or perceived to be] close to a win by the gambler) on gambling tasks.[124] Our results imply that BEA may

have a different molecular mechanism compared to SUD and that the brain regions around ACC, extending from vmPFC and MCC to SMA, underlie the distinction between SUD and BEA. These regions are connected by the occipitofrontal fasciculus shown in our results and directly affect anterior and posterior information communication as well as higher executive control and emotion-regulation functions.[122] It is reasonable to speculate that the impairment of these brain regions in BEA is less severe than that in SUD. This set of results highlights the importance of molecular-level information as markers in studying the neural mechanisms of addiction and in NIBS-based therapies.[125] In patients with IGD, addiction cues induce decreased functional connectivity from the ACC to the striatum.[126] This decrease is also associated with a number of comorbid symptoms, such as impaired executive control and symptoms of loneliness.[118] The response to treatment in subregions of ACC is also different. The intervention of prazepam for cocaine users can improve the response of dACC to an intra-abdominal pre-tied back cognitive task,[124] and similar results have been confirmed in heroin addiction.[127] While ACC is selected as the target area for intervention, the intervention plan should be considered.

We further explored the relationship between these ALFF differences and gene expression and receptor distribution. We used PLS to test the correlation between ALFF patterns and the expression levels of more than 10,000 genes and further assessed whether the gene expression pattern in relation to the density of addiction-related receptors such as DRD2, mu, DAT, and NAT is involved in the addiction process.[128,129] We found that a set of genes in SUD explain its abnormal activity patterns, whereas no significant functional enrichment of differentially related genes were found in BEA. Further functional labeling revealed that these genes may be related to the regulation of the dopamine pathway,[130] thereby affecting the distribution and activity of addiction-related receptors such as DRD2, opioid receptor mu 1 (OPRM1), DAT, and NAT. For SUD, addition-related genes are highly expressed in cortical glutamatergic,[131] midbrain dopaminergic,[132] GABAergic, and serotonergic neurons.[129] The development of new approaches for treating addiction through the study of receptor transporters has been a realm of interest for researchers.[133,134] Early studies have reported specific and robust changes in dopamine-rich areas of the cerebral cortex (e.g., ACC), with significantly reduced D1 receptor-G protein coupling, altered GABAergic systems, and cone dendritic overgrowth under exposure to cocaine.[135] In patients with opioid addiction, NMDA and glutamate/glutamine concentrations in dACC were decreased.[136] Gabapentin caused a more significant increase in glutamate in dACC and a greater decrease in GABA levels in patients with alcohol use disorder.[137] We also found that although the receptor distribution in BEA was also associated with multiple receptors/transporters, further gene enrichment did not support these interpretations. BEA had no clear functional enrichment of difference-related genes. These results suggest that the differences between SUD and BEA in ALFF may be related to the differential effects of these addiction types on the brain at functional and molecular levels, thus manifesting as different alterations in regional brain activity. The deviation in resting brain activity and the relationship between genes and receptors can be viewed as a dynamic interaction: abnormal gene expression brings about changes in receptor distribution, thereby increasing the susceptibility to addiction; conversely, addiction can excessively engage and alter the receptors in these brain areas, potentially influencing the functional and structural characteristics of the regions. These modifications might even extend to the realm of epigenetics,[138] further shaping the course of addiction and its associated neurological changes.

Conclusions

This study uncovered the common and distinct resting-state brain activity abnormalities in addiction, shedding new light on the neurobiological mechanisms of the disorder. The ACC, striatum, and SMA were identified as three key regions showing aberrant neural activity, and these may serve as potential targets for neuromodulation intervention.

Limitations of the study

The study has some limitations. First, we only used a CB meta-analysis method without considering the functional connectivity between brain regions. In particular, differences in small brain structures, such as those in subcortical areas, might be challenging to explore. Second, our meta-analysis was limited to resting-state fMRI studies using ReHo and/or ALFF without considering other possible imaging signatures. We excluded other analysis methods, such as independent component analysis, graph theory, and seed-based functional connectivity, since these measures could not be easily incorporated into the same meta-analysis framework. At the same time, we failed to select a more stringent threshold free cluster enhancement correction to reveal a particularly significant brain region, which may require more consistent addiction targets and indicators. Third, our data included participants from different withdrawal states and unbalanced sex distribution (fewer female participants; see Table 1). We included gender ratios in our analysis, and the results show that gender can have an impact as shown in a previous study,[139] with the main areas of influence being in frontal cortex and striatum (especially the caudate), as seen in Figure S3 and Table S6. Fourth, we included an imbalanced type of addiction studies, mostly alcohol use disorder and nicotine dependence. Therefore, future studies can further explore the resting-state brain activity characteristics of addiction by improving meta-analysis methods, increasing resting-state brain activity signatures, and expanding the total number of individuals analyzed.

STAR★Methods

Key resources table

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
Allen Human Brain Atlas	Allen Institute for Brain Science	https://help.brain-map.org/ ↗
Brain Genomics Superstruct Project V10	Harvard Dataverse	https://doi.org/10.7910/DVN/ILXIKS ↗
Software and Algorithms		
Anisotropic effect-size version of the Seed-based *d* Mapping software package 5.15	SDM Project	https://www.sdmproject.com/ ↗
Network Modification Tool 2.1	Amy Kuceyeski	https://github.com/kjamison/nemo ↗
Neurosynth	Tal Yarkoni	https://neurosynth.org/ ↗

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Python 3.9.5	Python Enhancement Proposals	https://www.python.org/ ↗
Brainsmash 0.11.0	Joshua B Burt	https://brainsmash.readthedocs.io/ ↗
MATLAB 2022a	MathWorks	https://www.mathworks.com/ ↗
Gene Ontology enRIchment anaLysis and visuaLizAtion tool	Eran Eden, Roy Navon	http://cbl-gorilla.cs.technion.ac.il/ ↗
JuSpace v 1.3	Juergen Dukart	https://github.com/juryxy/JuSpace ↗

Resource availability

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Ti-Fei Yuan (ytf0707@126.com ↗).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- All data in this study can be obtained online (https://osf.io/kctb7/ ↗).
- All code in this study can be obtained online (https://osf.io/kctb7/ ↗), and the analysis was pre-registered (https://osf.io/bngxd ↗).
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

Experimental model and study participant details

This study did not generate experimental model or enroll subjects. In this study, the coordinates data about the fMRI was obtained from previous studies.

Data sources

Meta-analyses datasets

Inclusion and exclusion criteria for selecting studies

Inclusion criteria for selecting studies are as follows: 1) peer-reviewed, original articles; 2) direct comparison between addiction patients and healthy controls (HCs); 3) a canonical diagnosis based on the DSM-IV, DSM-5, or other published scales with high reliability and validity; 4) whole-brain analysis of resting-state fMRI results that reported coordinates based on peak *t*, *p* or z-values of the ReHo or ALFF difference; and 5) articles written in English. Exclusion criteria are: 1) previous studies that used the same group of participants; or 2) other fundamental biological/clinical/mechanical issues that might affect the conclusions (e.g., control participants with other mental illnesses).

Database generation

Two authors (H-Z, X-L) independently searched PubMed (https://pubmed.ncbi.nlm.nih.gov/ ↗), Web of Science (http://apps.webofknowledge.com ↗), Cochrane library (http://www.cochranelibrary.com/ ↗), and OSF registries (http://osf.io/registries/ ↗) for articles published from January 1st, 2001 to Jun 15th, 2023 using the following search terms: ((ReHo OR "Regional Homogeneity") OR ("Amplitude of low-frequency fluctuation" OR "fractional Amplitude of low-frequency fluctuation" OR ALFF OR fALFF OR LFF) AND ("drug dependence" OR "drug use disorder" OR "Substance use disorder" OR ((alcohol OR nicotine OR smoke OR marijuana OR cannabis OR cocaine OR methamphetamine OR drug OR morphine OR heroin OR gaming OR Internet OR eating OR food OR gambling) AND addict *)).

The search was refined by "DOCUMENT TYPES: (ARTICLE) AND LANGUAGES: (ENGLISH)". Based on these search criteria, we created a database of references. In addition, studies were determined using bibliometric analysis software (HistCite) to locate the related articles. The two authors independently screened titles and abstracts and deleted duplicates and records that did not meet the inclusion criteria and/or meet the exclusion criteria. Full articles were then evaluated carefully by each author to produce separate inclusion lists. This procedure has complied with the preferred item selection method for Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020[140] in supplement TableS1 and is illustrated in web shinyapp (https://www.eshackathon.org/software/PRISMA2020.html ↗) (Figure1A). We modified a quality assessment checklist for articles applicable to fMRI studies of addictive disorders and assessed each included contrast, and all included studies were in the higher-quality supremacy (scores greater than 9), with specific results in the TableS2.

Method details

Meta-analysis process

Main meta-analysis process

The meta-analysis used the anisotropic effect-size version of the Seed-based *d* Mapping (AES-SDM) software package version 5.15 (http://www.sdmproject.com ↗). The AES-SDM is a reliable and effective voxel-based meta-analysis method published in high-quality journals.[141,142,143] This method combines hyperactivation and hypoactivation in one analysis, while other MRI meta-analysis approaches (e.g., ALE, MKDA) usually include only one side. In the subsequent analysis, we also performed separate statistics and analyses based on the reported increases and decreases in the original study.

Detailed analytical steps

Our meta-analysis was conducted in the following steps. 1) We prepared input for the AES-SDM software to collect raw information, and the main outcomes from the included studies were used to create the SDM table. 2) The Hedges' effect-size map and an effect-size variance map were recreated for comparing addiction patients and HCs from peak coordinates and effect sizes for each included study, using a nonnormative anisotropic Gaussian kernel.[32] A mean map was then made by a voxel-wise calculation using a random-effects model, weighted by sample size for each comparison and inter-study heterogeneity. Statistical significance was determined using standard permutation tests. We use the 20mm isotropic full width at half maximum (FWHM) in SDM according to previous studies.[144] Conventionally, if a study employs more than one comparison, only one would be selected for meta-analysis. However, with ES-SDM 5.15, we can combine the results of the iterative measurements to achieve more reliable results.[141] 3) A mean comparison of resting brain activity (ReHo/ALFF) regions between addiction populations and HCs was performed. 4) A subgroup analysis was performed for two resting-state fMRI measurements: ReHo and ALFF/fALFF. 5) Conjunctional (Multimodal) analysis was applied to examine the convergence between the subgroup analysis (ReHo and ALFF/fALFF).[145] 6) Comparison between two groups in the ALFF index is then performed with linear model analyses. 7) We conducted a jackknife sensitivity analysis by removing each study to assess the robustness of the results. 8) The statistical heterogeneity of individual clusters was examined using a random-effect model.

Based on the empirical validation in the previous study,[32] we selected the uncorrected p value of 0.005 as the primary threshold, as this was found to be an optimal balance between sensitivity and specificity and also to be approximately equivalent to the corrected p value of 0.05 (indeed p value of 0.025 for each one tail) in ES-SDM. Other than the p value, our analysis was set cluster voxel number ≥50. For each significant addiction case-control comparison, Egger's test was used to assess the asymmetry of the funnel plot as a measure of potential publication bias.[144] Because the ALFF results of BEA do not cover the whole brain (78/268) the subsequent PLS and PET correlations results will be affected by this fact. We performed subgroup analyses of some of the factors that might affect the results, and for this purpose, we included only 3T MRI, nicotine-dependent patients, patients with online gaming disorders, fALFF, and ALFF. We did subgroup analyses, respectively, using the sex ratio as a meta-regression, and found that there would be heterogeneity in some of the brain regions. The results mentioned above are shown in the Supplementary Note.

The robust analysis

Although the ReHo and ALFF methods are mathematically different, both methods measure the local fluctuation of spontaneous brain activity. A previous study of attention deficit hyperactivity disorder found that ReHo and ALFF have convergent regions showing abnormalities.[146] Therefore, we did a conjunction analysis by combining the meta-analysis of ReHo and the meta-analysis of ALFF. The following analyses were conducted to measure the difference between SUD and BEA (FiguresS4, S5, and TableS4). For post hoc analysis of the reconstructed T maps, the reconstructed variance and effect size were extracted for correlation analysis based on the significantly different brain regions as a mask (Figure2C) for independent samples t-test.

To further compare the difference between SUD and BEA, and at the same time, due to the limitation of sample size (ReHo articles only three contrasts between BEA and HC), we only use the ALFF in the

subgroup analyses and follow-up analyses (describe in the following sections). We compare the behavior addition patients (7 contrasts) and substance use disorder patients (22 contrasts) fALFF/ALFF by the linear model. The batch code is '*substance-behavior= lm substacne*'. Applying meta-regression analysis to predict brain impairment using addiction time, cigarette number per day, impulsivity, anxiety, and depression difference (statistical t value), respectively (Radua etal., 2014). The batch code is '*Regression-druguse time= lm Addiction_year*'.

To check the robustness of these two meta-analyses, we use Egger's test to explore potential publication bias, the I2 and Q to test the homogeneity, and the jackknife analysis to test the single study weight of the whole meta-analysis. In addition, we also summarize the questionnaire about impulsivity, anxiety, and depression of recent meta-analyses.

Publication bias and robustness analysis

Egger's test did not reveal publication bias for the ReHo meta-analysis. But there was a publication bias in the right postcentral gyrus for ReHo (TableS4 and FigureS1). Jackknife sensitivity analyses indicated that the results of ReHo and ALFF are robust (TableS4). Specially, the jackknife sensitivity analyses were performed by sequentially deleting the included studies and comparing the consistency of the post-deletion results with the pre-deletion results, and both ReHo (23/26) and ALFF (27/29) showed extremely high stability, except for ALFF which would have been left with abnormally elevated results in the right striatum in some cases, suggesting that the right striatum may be an important brain region. The heterogeneity analysis showed higher heterogeneity in the corpus callosum, MCC, right middle occipital gyrus, left precentral gyrus, right anterior thalamic projections, right postcentral gyrus, left superior frontal gyrus, medial, left PCC, right striatum, and left amygdala (TableS5 and FigureS1).

Association with the probability of white matter

This study also used the Network Modification (NeMo) tool[147] to infer the structural connectivity network that may underlie the changes in spontaneous brain activity in this study. Based on structural connectivity mapping in 73 healthy subjects, we normalized brain changes obtained from meta-analysis to a common MNI space, first using linear kernels and then non-linear normalization in SPM8. The Connectivity Change (ChaCo) score for the percentage of pathways in each GM region that connects to the other areas and passes through the WM anomaly mask. For example, if a GM region has 1000 connected WM bundles and 50 passes through any voxel in the WM anomaly mask, then the ChaCo score for that region would be 50/1000= 5%, or −0.05 (a negative number indicates a loss of connectivity).

Association between cognitive terms

We investigated the relationship between meta-analytic cognitive concepts and addiction-related ALFF/ReHo abnormalities using Neurosynth (https://neurosynth.org/ ↗).[35] The thresholded Z-maps resulting from the group comparisons of regional ALFF/ReHo scores were separated into positive (addiction>controls) and negative (addiction<controls) maps. We examined the spatial connection between these positive- or negative-maps and the meta-analytic map of each term in the database using Neurosynth's "decoder" function. Permutation tests with adjusted spatial autocorrelations were used to determine the significance of the correlation coefficients (https://brainsmash.readthedocs.io/ ↗).[148] The correlation with the top 10 cognitive or full 10 anatomy terms

was thresholded and visualized at r>0.10 (Figures2D and 5D).

Association between gene expression

We investigated the relationship between transcriptional patterns and between-group differences in the main ALFF/ReHo map using a partial least squares (PLS) regression analysis. We first used a cortical parcellation atlas to align the gene expression data (15,633 genes) and between-group differential Z-map of the main ALFF/ReHo. We used the python package abagen (abagen.readthedocs.io) which provides a reproducible workflow for processing and preparing the Allen Human Brain Atlas (AHBA, https://help.brain-map.org/ ↗) microarray expression data for analysis. According to the standardized processing procedure we followed an important consideration regarding sample assignment is that only two out of six brains were sampled from both hemispheres and four brains had samples collected only from the left. This sparse sampling should be carefully considered when combining data across donors. Restricting analyses to the left hemisphere would minimize variability across regions (and hemispheres), as well as the number of samples available. Accordingly, there is an option in the toolbox to mirror samples across left-right hemispheres. However, given the asymmetry we found in the results of brain regions in addictive disorders, we included both hemispheres, i.e., we had data from six donors on the left side and two donors on the right side. The predictor variables were gene expression data, and the response variable was the Z-maps of the between-group differences of the major ALFF/ReHo, respectively. To see if the R^2 of the PLS component was significantly greater than that expected by chance, we used a spatial autocorrelation-adjusted permutation test. The importance of the gene was measured by the Variable Importance in Projection (VIP) score, estimated with the PLS regression model. The VIP score greater than one is generally used as a criterion for detecting the relative importance of an indicator. We used a bootstrapping method to correct each gene's weight estimation error for each significant component.[33] The genes were then sorted based on their adjusted weight, which reflects their contribution to the PLS regression component.

Gorilla (http://cbl-gorilla.cs.technion.ac.il/ ↗)[149] was used to find enriched Gene Ontology concepts by including both the descending and ascending sequences in the gene enrichment analysis. The biological process, molecular function, and cellular component ontology categories were all evaluated. Benjamini-Hochberg false discovery rate (FDR)-corrected q<0.05 was used to define significant enrichment.[150] In this connectome-transcriptome association analysis, we verified the sex and hemisphere effects. We investigated the disorder-related genes from the high-resolution *in situ* hybridization image data from the AHBA to see if they contributed more to the PLS model than other disorder-related genes reported in earlier gene studies. We chose genes associated with addiction and six other brain illnesses from the gene list. Using permutation tests, we compared the PLS weight of addiction genes to those of other illnesses. We also examined the relationship between each gene's expression profile and the between-group Z-map of the main ALFF/ReHo. Permutation tests were used to investigate the percentage of significantly linked genes between addiction and other illnesses.

Association between the distribution of receptors/transporters

Finally, we evaluated the topographic relationship between the abnormal spontaneous activation and the distribution of several receptor/transporter systems by assessing the identified receptors/transporters in abnormal spontaneously activated brain regions relative to the whole brain using the

JuSpace v1.3 (https://github.com/juryxy/JuSpace ↗). The averaged receptor/transporter density of all nodes within a given network was compared with a null distribution based on 10,000 random network configurations. We selected 5-hydroxytryptamine (5-HT1A, 5-HT2A), dopamine (D2/D3), and gabalaminobutyric acid (GABA) receptors, as well as F-DOPA (reflecting presynaptic dopamine synthesis capacity) and N-acetyltransferase and Vesicular acetylcholine transporter. Although all systems have been associated with addiction through previous studies, we use the average group maps scanned in a multivector molecular imaging atlas in healthy human populations. For comparison, these maps, in MNI152 space, were resampled to an isotropic 2mm spatial resolution, as in our fMRI data, and were linearly rescaled to the range of 0–100. For computational efficiency, only those significantly higher expressed receptors/transporters were entered into the spatial correlation analysis. Brain regions were reassigned to 119 brain regions (following the MICCAI 2012 Grand Challenge and Workshop on Multi-Atlas Labeling atlas). To ensure robustness, the volumetric results of brain regions were used as covariates. To determine the statistical significance of the spatial correlation, a permutation test (10,000 times) was used.[151] In addition, during the comparison of SUD and BEA, we compared the difference in brain maps using model 2 (two images compare model), with±0.241 as a meaningful R value ($p<0.05/6$) shown in the FigureS5.

Activation network mapping

To test whether the differences in our meta-analysis are localized on a shared network, we used a new technique called abnormal spontaneous Activation Network Mapping (ANM),[152] which extended from the Lesion Network Mapping (LNM) technique.[153] We generated abnormal spontaneous activation masks individually for each contrast (n= 50) from the previous meta-analysis and used them as seed regions for connectivity analysis. The normative connectome of 1000 subjects were calculated based on Genome Superstruct Project (GSP).[154] Specifically, for each normal subject in our GSP dataset, we calculated Pearson correlation coefficients between the mean time course of all voxels within the abnormal spontaneous activity masks and the time course of each voxel in the whole brain. The resulting 1,000-subject-level r maps were converted to Fisher's z maps by Fisher's z transformation.

We also conducted sensitivity analyses using two methods.[152] In method 1, the above 1,000 subject-level Fisher z mappings for each experiment were averaged to create an experiment level average Fisher z plot. These experiment-level averaged Fisher z mappings belonging to the same group (e.g., the same resting-state index category) were then compared to zero in one sample t-test with a cluster-level FWE $p<0.05$ (cluster-level formation threshold $p<0.001$) when corrected for multiple comparisons. For groups of more than 100 experiments, the threshold for group-level t-plots was a more conservative $t>8$ (equivalent to $p<0.001$ for the voxel method correction). Groups above the threshold were across brain regions significantly associated with activation masks in the experiments. In method 2, the above 1,000 subject-level Fisher z mappings for each experiment was compared to zero using a voxel-wise one-sample t test. The t-maps for each experiment level were then threshold and binarized at $t>8$ (corresponding to $p<0.001$ for voxel wise family error (FWE) correction). Finally, all binarized maps were overlaid and threshold at 60% to create a group-level activation network overlay map. The suprathreshold clusters in this map were brain regions that were functionally associated with more than 60% (30/50) of the activation masks. The two methods would reveal two different potential functional networks. In particular, we make some adjustments to the original method. For statistical networks we define $z>0$ or $t>8$ as positive connection

networks. The networks with z<0 or t<−8 are defined as negatively connected networks.

In addition, to serve as a benchmark for revealing addiction-related abnormal spontaneous activation, we employed Standard voxel-wise lesion-symptom mapping (VLSM) for testing lesion location and clinical outcomes (years of addiction) using NPM software (2MAY2016 64bit). Study site and lesion size were included as covariates, and abnormal voxels associated with ReHo (n= 22) versus ALFF (n= 28) were identified.

Quantification and statistical analysis

All the statistical software and content involved is detailed in each section of method details section.

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Author contributions

H.Z., T.Z., L.X., G.D., Y.Y., and T.-F.Y. designed the overall experiments and had unrestricted access to all data. H.Z., T.Z., L.X., G.D., Y.Y., and T.-F.Y. performed statistical analyses and drafted the manuscript. G.D., Y.Y., and T.-F.Y. provided critical reading of the manuscript and insightful comments. All authors read and approved the final draft of the manuscript and take full responsibility for its content.

Declaration of interests

The authors declare no competing interests.

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